Solutions to Chapter 10 Exercises

Chaos: An Introduction to Dynamical Systems (Alligood, Sauer, Yorke)

Definitions used. For a hyperbolic fixed point p of an invertible map f, $\mathcal{S}(p) = \{x : |f^n(x) - p| \to 0 \text{ as } n \to \infty\}$ and $\mathcal{U}(p) = \{x : |f^{-n}(x) - p| \to 0 \text{ as } n \to \infty\}$. A point $x \neq p$ in $\mathcal{S}(p) \cap \mathcal{U}(p)$ is a homoclinic point; it is transverse if the tangent lines of the two manifolds at x are distinct.

Exercise 10.1

Let p be a fixed point and f a diffeomorphism. If $x \in \mathcal{S}(p)$, then $|f^n(f(x)) - p| = |f^{n+1}(x) - p| \to 0$, so $f(x) \in \mathcal{S}(p)$; thus $f(\mathcal{S}) \subseteq \mathcal{S}$. Conversely, if $y \in \mathcal{S}$ and $x = f^{-1}(y)$, then $|f^{n+1}(x) - p| = |f^n(y) - p| \to 0$, so $x \in \mathcal{S}$; hence $f^{-1}(\mathcal{S}) \subseteq \mathcal{S}$ and $f(\mathcal{S}) = \mathcal{S}$. The same argument applied to f^{-1} gives $f(\mathcal{U}) = \mathcal{U}$. (For a periodic orbit $\{p_0, \ldots, p_{k-1}\}$ the same computation shows $f(\mathcal{S}(p_i)) = \mathcal{S}(p_{i+1})$, so the union over the orbit is invariant.)

Exercise 10.2

Let A be an invertible 2×2 matrix and write $\mathbb{R}^2 = E^s \oplus E^c \oplus E^u$ for the sums of generalized eigenspaces for eigenvalues with $|\lambda| < 1$, $= 1$, > 1 (complex pairs have equal moduli, so they go together). For $v = v_s + v_c + v_u$, $A^n v \to 0$ iff $v_c = v_u = 0$ (on E^c the iterates $A^n v_c$ stay bounded away from 0; on E^u they grow). So $\mathcal{S}(0) = E^s$, and applying the same to A^{-1}, $\mathcal{U}(0) = E^u$. The possibilities are:

- both $|\lambda| < 1$ (sink): $\mathcal{S} = \mathbb{R}^2$, $\mathcal{U} = \{0\}$;
- both $|\lambda| > 1$ (source): $\mathcal{S} = \{0\}$, $\mathcal{U} = \mathbb{R}^2$;
- $|s| < 1 < |u|$ (saddle): $\mathcal{S}$ and $\mathcal{U}$ are the two lines along the eigenvectors of s and u;
- $|\lambda_1| < 1 = |\lambda_2|$: $\mathcal{S}$ is a line, $\mathcal{U} = \{0\}$;
- $|\lambda_1| > 1 = |\lambda_2|$: $\mathcal{S} = \{0\}$, $\mathcal{U}$ is a line;
- both $|\lambda| = 1$ (rotation, shear, $\pm I$, $\ldots$): $\mathcal{S} = \mathcal{U} = \{0\}$.

Exercise 10.3

Let p be the saddle.

(a) Let x be a transverse homoclinic point. By Exercise 10.1, $f(x) \in \mathcal{S} \cap \mathcal{U}$, and $f(x) \neq p$ since f is one-to-one with $f(p) = p$; so $f(x)$ is a homoclinic point. Moreover f maps $\mathcal{S}$ onto $\mathcal{S}$ and $\mathcal{U}$ onto $\mathcal{U}$, so the invertible linear map $Df(x)$ carries the tangent line $T_x\mathcal{S}$ onto $T_{f(x)}\mathcal{S}$ and $T_x\mathcal{U}$ onto $T_{f(x)}\mathcal{U}$. Since $T_x\mathcal{S} \neq T_x\mathcal{U}$ and $Df(x)$ is invertible, $T_{f(x)}\mathcal{S} \neq T_{f(x)}\mathcal{U}$: the crossing at $f(x)$ is transverse. The same argument with f^{-1} gives $f^{-1}(x)$.

(b) Take the transverse crossing x_1 of Figure 10.12 (the vertical branch of $\mathcal{U}$ leaving p loops over and crosses the horizontal branch of $\mathcal{S}$ at x_1). Forward iterates $f^k(x_1)$, $k = 1, 2, 3$, lie on $\mathcal{S}$ between x_1 and p, getting closer to p (points of $\mathcal{S}$ move toward p). They also lie on $\mathcal{U}$, so $\mathcal{U}$ must pass through each of them: the loop of $\mathcal{U}$ is pushed along $\mathcal{U}$ and forms three further thinner and thinner loops crossing the horizontal branch at $f(x_1), f^2(x_1), f^3(x_1)$, accumulating toward p. Backward iterates $f^{-k}(x_1)$, $k = 1, 2, 3$, lie on $\mathcal{U}$ between p and x_1 (points of $\mathcal{U}$ move toward p under f^{-1}), and on $\mathcal{S}$ farther from p than x_1, successively farther out along the horizontal branch. So $\mathcal{S}$ must extend to pass through them: it forms three loops on the far side, crossing the vertical branch of $\mathcal{U}$ near p at $f^{-1}(x_1), f^{-2}(x_1), f^{-3}(x_1)$. Each new crossing is again transverse by (a), and the arc of a manifold between $f^k(x_1)$ and $f^{k+1}(x_1)$ is the image under f^k of the arc between x_1 and $f(x_1)$.

Exercise 10.4

$f(x, y) = (x/2,\ 2y - 7x^2)$. For a point $(x, 4x^2)$:

$$f(x, 4x^2) = \left(\tfrac{x}{2},\ 8x^2 - 7x^2\right) = \left(\tfrac{x}{2},\ x^2\right) = \left(\tfrac{x}{2},\ 4(\tfrac{x}{2})^2\right),$$

so the parabola $\Gamma = \{(x, 4x^2)\}$ is invariant and $f^n(x, 4x^2) = (x/2^n,\ 4x^2/4^n) \to (0, 0)$. Thus $\Gamma \subseteq \mathcal{S}(0, 0)$.

Why Γ is the whole stable manifold: $Df(0, 0) = \begin{pmatrix} 1/2 & 0 \\ 0 & 2 \end{pmatrix}$, so $(0, 0)$ is a saddle and by the Stable Manifold Theorem $\mathcal{S}(0, 0)$ is a single one-dimensional curve through the origin. Γ is already a one-dimensional curve through the origin with no endpoints, so it cannot be a proper part of $\mathcal{S}$ (the stable manifold is a curve with no endpoints, by 10.6). One can also verify directly: put $w = y - 4x^2$. Then $w' = 2y - 7x^2 - 4(x/2)^2 = 2(y - 4x^2) = 2w$, so for $w_0 \neq 0$, $w_n = 2^n w_0$ while $x_n \to 0$, so $y_n \to \pm\infty$ and the orbit does not converge to the origin. Hence $\mathcal{S} = \Gamma$.

Exercise 10.5

$f(x, y) = (x^2 - 5x + y,\ x^2)$. Fixed points: $y = x^2$ and $x = 2x^2 - 5x$, so $2x^2 - 6x = 0$: $(0, 0)$ and $(3, 9)$. $Df = \begin{pmatrix} 2x - 5 & 1 \\ 2x & 0 \end{pmatrix}$.

- $(0, 0)$: $Df = \begin{pmatrix} -5 & 1 \\ 0 & 0 \end{pmatrix}$ has eigenvalues -5 (eigenvector $(1, 0)$) and 0 (eigenvector $(1, 5)$). So it is a saddle: there is a stable manifold tangent to $(1, 5)$ (this direction is collapsed to the origin to first order) and an unstable manifold tangent to $(1, 0)$. Near $(0, 0)$ points on the unstable side are expanded by about 5 and flipped to the opposite side of $(0, 0)$ at each step (negative eigenvalue). (Caveat: $\det Df(0, 0) = 0$, so f is not a local diffeomorphism there;

the local stable and unstable manifold statements still hold for the hyperbolic fixed point of a C^1 map, but the global description via f^{-1} in Theorem 10.1 does not apply as stated.)

- $(3,9)$: $Df = \begin{pmatrix} 1 & 1 \\ 6 & 0 \end{pmatrix}$ has eigenvalues 3 (eigenvector $(1,2)$) and -2 (eigenvector $(1,-3)$). Both have modulus > 1, so it is a source: $\mathcal{S} = \{(3,9)\}$ and $\mathcal{U}$ contains a neighborhood of it. Nearby points move away, stretched by 3 along $(1,2)$ and by 2 along $(1,-3)$ with a flip to the other side.

Exercise 10.6

Let p be a saddle of the invertible planar map f. By Theorem 10.1, $\mathcal{S}(p)$ is a one-dimensional manifold containing p in its interior; choose a small open arc $\mathcal{S}_\varepsilon \subset \mathcal{S}$ around p (an injective image of an open interval) with $f(\mathcal{S}_\varepsilon) \subseteq \mathcal{S}_\varepsilon$, which is possible because f contracts along $\mathcal{S}$ near p. Every $x \in \mathcal{S}$ has $f^n(x) \in \mathcal{S}_\varepsilon$ for large n, so

$$\mathcal{S} = \bigcup_{n \geq 0} f^{-n}(\mathcal{S}_\varepsilon),$$

an increasing union ($f^{-n}(\mathcal{S}_\varepsilon) \subseteq f^{-(n+1)}(\mathcal{S}_\varepsilon)$). Each $f^{-n}(\mathcal{S}_\varepsilon)$ is the image of an open arc under the homeomorphism f^{-n}, hence an open arc without endpoints that does not cross itself (injectivity). An increasing union of open embedded arcs, each extending the previous one, has no endpoints and no self-crossings (a self-crossing would already occur on one of the arcs, contradicting injectivity of f^{-n} on $\mathcal{S}_\varepsilon$). The same argument with f^{-1} applies to $\mathcal{U}(p)$.

Exercise 10.7

The map is $v \mapsto Av \pmod 1$ with $A = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}$, $\det A = 1$, $\operatorname{tr} A = 3$. The eigenvalues are $\lambda_u = \frac{3+\sqrt{5}}{2} > 1$ and $\lambda_s = \frac{3-\sqrt{5}}{2} < 1$, with eigenvectors $(1, \varphi)$ and $(1, -1/\varphi)$, where $\varphi = \frac{1+\sqrt{5}}{2}$.

Unstable manifold. $\mathcal{U}(0,0) = \{(t, \varphi t) \bmod 1 : t \in \mathbb{R}\}$: the line of slope φ through the origin, wrapped around the torus. Indeed $A^{-n}(t, \varphi t) = \lambda_u^{-n}(t, \varphi t) \to 0$. Conversely, if $f^{-n}(v) \to 0$, lift v to $\tilde{v} \in \mathbb{R}^2$ and choose integer vectors k_n with $A^{-n}\tilde{v} - k_n \to 0$. Then $k_{n+1} - A^{-1}k_n$ is an integer vector (because A^{-1} has integer entries) tending to 0, so it is 0 for large n. Hence for some N and all $m \geq 0$, $A^{-m}w \to 0$ with $w = A^{-N}\tilde{v} - k_N$. Since A^{-1} expands the λ_s-eigendirection, w must lie on the λ_u-eigenline, and $v \equiv A^N w \pmod 1$ lies on that line's image. Because the slope φ is irrational, $\mathcal{U}$ winds densely around the torus.

Stable manifold. In the same way $\mathcal{S}(0,0) = \{(t, -t/\varphi) \bmod 1\}$, a line of irrational slope $-1/\varphi = \frac{1-\sqrt{5}}{2}$, also dense.

Right angles. Both manifolds are straight lines (in the universal cover) with constant direction vectors $(1, \varphi)$ and $(1, -1/\varphi)$, and

$$(1, \varphi) \cdot (1, -1/\varphi) = 1 - 1 = 0.$$

So at every point where they meet, in particular at the origin and at all the infinitely many homoclinic points, they cross at a right angle (as expected, since A is symmetric and its eigenvectors are orthogonal). All crossings are transverse.

Exercise 10.8

Stable Manifold Theorem for $\mathbb{R}^n$. Let f be a diffeomorphism of $\mathbb{R}^n$ and let p be a hyperbolic fixed point, i.e. $Df(p)$ has no eigenvalue of modulus 1. Let E^s be the span of the generalized eigenvectors for eigenvalues of modulus < 1, of dimension k (the number of such eigenvalues counted with multiplicity), and E^u the span of those for eigenvalues of modulus > 1, of dimension $n-k$. Then the stable manifold $\mathcal{S}(p)$ is a k-dimensional smooth (injectively immersed) manifold containing p and tangent to E^s at p, and the unstable manifold $\mathcal{U}(p)$ is an $(n-k)$-dimensional smooth manifold containing p and tangent to E^u at p; both are invariant under f. The corresponding statement holds for a hyperbolic periodic point of period m using Df^m, with a stable and unstable manifold at each point of the orbit.

Exercise 10.9

I take the typo in the statement to mean $f(x,y) = (x/3,\, 3y)$ for $0 \le y \le \frac{1}{3}$ and $f(x,y) = (1 - x/3,\, 3 - 3y)$ for $\frac{2}{3} \le y \le 1$. The strips $H_0 = [0,1] \times [0, \frac{1}{3}]$ and $H_1 = [0,1] \times [\frac{2}{3}, 1]$ are mapped onto the vertical strips $V_0 = [0, \frac{1}{3}] \times [0,1]$ and $V_1 = [\frac{2}{3}, 1] \times [0,1]$, and the middle strip is mapped outside the square (the "cap"). The origin is a fixed saddle with $Df(0,0) = \mathrm{diag}(\frac{1}{3}, 3)$. Let

$$E = \{0,\, 1,\, \tfrac{1}{3},\, \tfrac{2}{3},\, \tfrac{1}{9},\, \tfrac{2}{9},\, \tfrac{7}{9},\, \tfrac{8}{9}, \dots\}$$

be the endpoints of the intervals in the construction of the middle-thirds Cantor set (the points of the Cantor set that are eventually mapped to 0 by $y \mapsto 3y$ on $[0, \frac{1}{3}]$, $y \mapsto 3 - 3y$ on $[\frac{2}{3}, 1]$). Then

$$\mathcal{S}(0,0) \cap [0,1]^2 = [0,1] \times E, \qquad \mathcal{U}(0,0) \cap [0,1]^2 = E \times [0,1],$$

that is, countably many horizontal segments and vertical segments, respectively.

Stable manifold. If the orbit of (x,y) stays in the square and tends to $(0,0)$, then the y-coordinates follow the expanding map $y \mapsto 3y$ or $3-3y$ and must tend to 0, which is possible only if some $y_n = 0$ exactly; that is, $y \in E$. Conversely, if $y \in E$ then eventually $y_n = 0$ and then $(x_n, 0) \mapsto (x_n/3, 0) \to (0,0)$. So $\mathcal{S} \cap [0,1]^2 = [0,1] \times E$. Concretely: $y = 0$ (the bottom edge, since $f(x,0) = (x/3, 0)$); $y = 1$ (since $f(x,1) = (1 - x/3, 0)$); $y = \frac{1}{3}, \frac{2}{3}$ (mapped to $y = 1$); then $y = \frac{1}{9}, \frac{2}{9}, \frac{7}{9}, \frac{8}{9}$; and so on.

Unstable manifold. By symmetry (use f^{-1}, which on $V_0 \cup V_1$ is $(x,y) \mapsto (3x, y/3)$ resp. $(3(1 - x),\, 1 - y/3)$), the backward orbit tends to $(0,0)$ iff $x \in E$. Concretely: the segment $\{0\} \times [0, \frac{1}{3}]$ is mapped onto $\{0\} \times [0,1]$ (the left edge); f maps the left edge's part in H_1 onto the right edge $\{1\} \times [0,1]$; f maps the right edge into the vertical lines $x = \frac{1}{3}$ and $x = \frac{2}{3}$; these in turn into $x = \frac{1}{9}, \frac{2}{9}, \frac{7}{9}, \frac{8}{9}$; and so on, each segment $\{c\} \times [0,1]$ with $c \in E$ being joined to the next through the part of $\mathcal{U}$ in the cap outside the square.

Therefore the stable and unstable manifolds cross exactly at the points $E \times E \setminus \{(0,0)\}$ (for example $(1,0)$, $(0,1)$, $(\frac{1}{3}, \frac{2}{3})$, $\dots$), infinitely many transverse homoclinic points, each crossing being at a right angle.